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Digital Image Processing
Department of Computer Science
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Class Project
Leaf Classification using Fourier Descriptors

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Introduction

Shape is one of the prominent low level image features. We tend to analyze object properties by identifying their shapes. In this project, the objective is to classify leaves.
To do this we need two things: 1) a model to represent a particular class of leaves and 2) a classifier to assign a test example to a model. Many methods have been proposed using different feature sets and shape signatures. Out of which, methods based on Fourier descriptors achieved more prominence. In this project we divided feature extraction into four main categories. First one being shape signatures like Central distances (also called as CCDC), cumulative angular function, and curvature. Second one is the color features of the leaves like mean, standard deviation, skew-ness, and kurtosis. The third is texture features like fractal and lacunarity. And the fourth is a set of descriptors which can be called as global descriptors like Eccentricity, Aspect Ratio, Rectangularity, Dispersion and Roundedness. We used Dynamic-Time-Wrapping to train the model and classify the test leaf.

Preprocessing

We used Bi-modal Histogram based thresholding to binarize given leaf Image. Then we smoothed the binary image using Open-Close operation with a window size of three. We removed large holes using blobcoloring.

Contour Detection and Feature Extraction

After smoothing the given leaf image, we extract the contour of the image by doing a XOR operation between eroded image and the binary image.

Feature Extraction

1) **Shape Signatures**
   a) **Central distance (CCDC)**: The central distance is expressed by the distance between boundary points and centroid of the leaf.

   $$r(t) = \left(\left[x(t) - x_c \right]^2 + \left[y(t) - y_c \right]^2 \right)^{\frac{1}{2}}$$

   where (x_c, y_c) is the centroid of the shape.

   This feature is invariant to translation as we subtracted the centroid from each boundary pixel.

   b) **Curvature**: Curvature function defined in this project is defined as the differentiation of successive boundary angles.

   $$\theta(t) = \arctan \frac{y(t) - y(t-w)}{x(t) - x(t-w)}$$

   where w is the window size.

   This feature is invariant under translation and rotation.

c) **Cumulative Angular Function**: We used Cumulative angular function developed by Zahn and Roskies which is net amount of angular bend between starting position $z(0)$ and position $z(t)$ on the shape boundary.

$$\varphi(t) = \left[\theta(t) - \theta(0)\right] \mathrm{mod}(2\pi)$$

2) Color Features:

We calculated features like mean, standard deviation, skewness and kurtosis for each component of the R-G-B color space.

Mean $$\mu = \frac{1}{MN}\sum_{i=1}^{M}\sum_{j=1}^{N} P_{ij}$$

Standard deviation $$\sigma = \sqrt{\frac{1}{MN}\sum_{i=1}^{M}\sum_{j=1}^{N}(P_{ij} - \mu)^2}$$

Skewness $$\theta = \frac{\sum_{i=1}^{M}\sum_{j=1}^{N}(P_{ij} - \mu)^3}{MN\sigma^3}$$

Kurtosis $$\gamma = \frac{\sum_{i=1}^{M}\sum_{j=1}^{N}(P_{ij} - \mu)^4}{MN\sigma^4}$$

3) Texture Features:

a) We calculated fractal dimension of the leaf using Box-counting procedure. But we noticed that the fractal dimensions of leaves of different classes are almost the same. So in the dataset given for the project, inter class variance cannot be achieved with fractal dimension.

b) We calculated one more feature which is similar to lacunarity we took the log ratio of Image moment one and two. Even this didn't any good information through which we can classify the leaves.

4) Global Descriptors

a) Eccentricity: Eccentricity is a measure of how much the given leaf deviates from being circular. We calculated eccentricity by calculating the covariance matrix of the leaf and taking the first two eigen-values. Eccentricity is given by

$$E = \sqrt{1 - \frac{\lambda_2}{\lambda_1}} \text{ where } \lambda_1 < \lambda_2$$

b) Aspect Ratio: Aspect ratio (also called slimness) is the ratio of width of the leaf to length of the leaf.

$$AR = \frac{\lambda_1}{\lambda_2}$$

c) Rectangularity: Rectangularity is given by

$$R = \frac{(\max(x) - \min(x)) * (\max(y) - \min(y))}{\lambda_1 * \lambda_2}$$

where x and y are boundary coordinates.

d) Dispersion: Dispersion (also called irregularity) is defined as the ratio between the radius of maximum circle enclosing region and the minimum circle that can be contained in the region. This measure will increase as the region spreads.

$$D = \frac{\max\left(\sqrt{(x_i - \bar{x})^2 + (y_i - \bar{y})^2}\right)}{\min\left(\sqrt{(x_i - \bar{x})^2 + (y_i - \bar{y})^2}\right)}$$

where $\bar{x}$ & $\bar{y}$ are the centroid in x and y, x_i and y_i are contour pixel coordinates.

e) Roundedness: It is the ratio of the area to square of the perimeter.

$$\text{Roundedness} = \frac{4\pi A}{P^2}$$

where A is the area of the leaf and P is the perimeter of the leaf.

Shape indexing using Fourier Descriptors

Each shape signature is a 1-d array and we take the Fourier transform of this array. We considered the first 64 Fourier coefficients to index our shape model.

1) As the CCDC and curvature functions are not scale-invariant we divide the first 64 Fourier coefficients obtained by the DC component.

$$\mathbf{f} = \frac{|FD_1|}{|FD_0|}, \frac{|FD_2|}{|FD_0|}, \frac{|FD_3|}{|FD_0|}, \frac{|FD_4|}{|FD_0|} \ldots\ldots\ldots, \frac{\left|FD_{\frac{N}{2}}\right|}{|FD_0|}$$

2) As the cumulative angular function is invariant under translations, rotation and scales no other normalization is required.

Classification using Dynamic-Time-Wrapping

Dynamic time wrapping is an algorithm which is usually used for measuring similarity between two sequences which may vary in time or speed. It's a difference in distance based similarity measurement. In our project the DTW measures the similarity between the leaf feature-model matrices and test leaf's feature model matrix and assigns the leaf to least distance model.

Results

Out of ten test images, nine of them were classified correctly. When looking at the first leaf that was misclassified, we found that the training set for the leaf type varied considerably so much so that even we would have trouble classifying it correctly by eye.

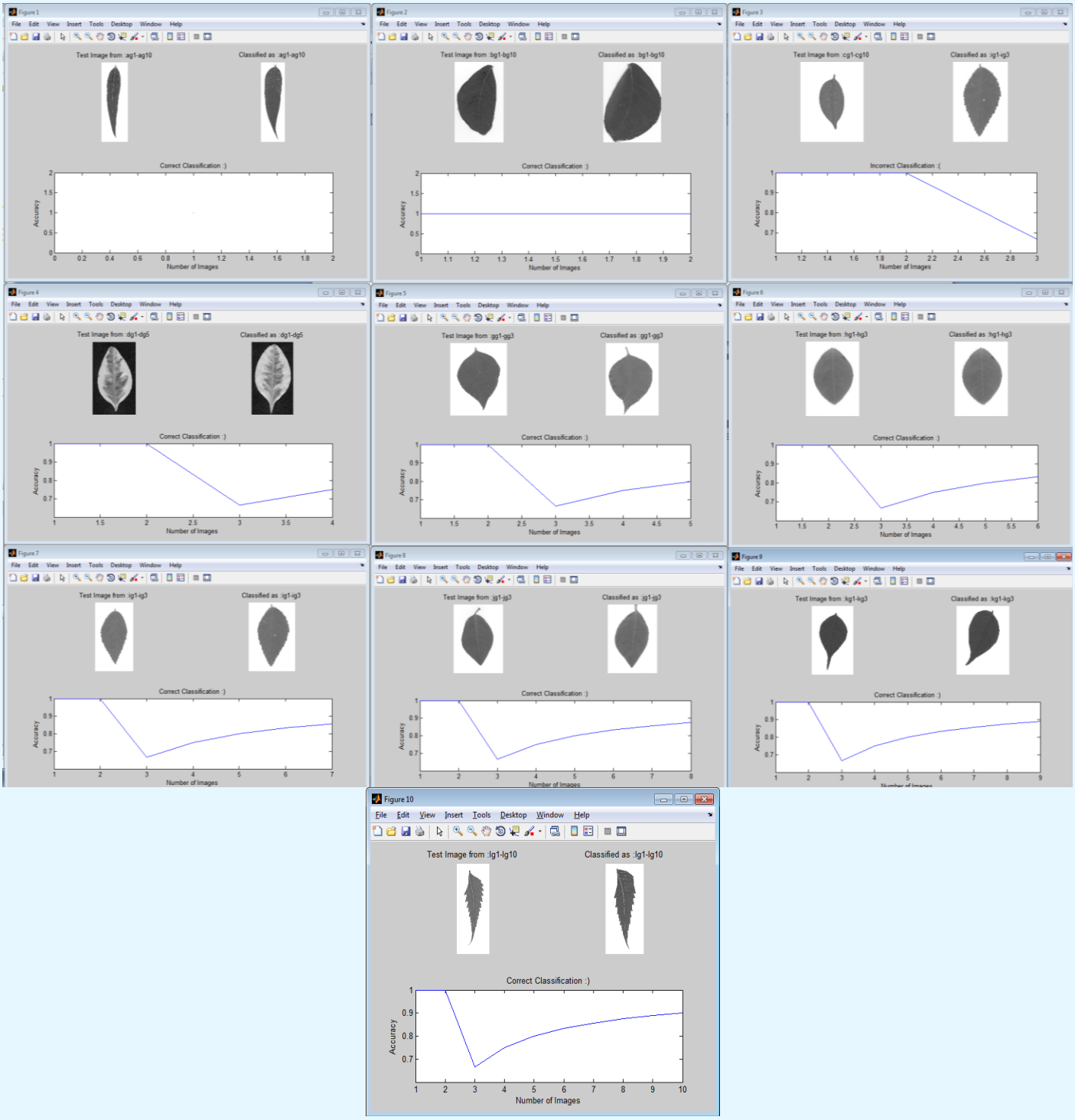
Figure 1
Test Image from :ag1-ag10
Classified as :ag1-ag10
Correct Classification :)
Accuracy
Number of Images
Figure 2
Test Image from :bg1-bg10
Classified as :bg1-bg10
Correct Classification :)
Accuracy
Number of Images
Figure 3
Test Image from :cg1-cg10
Classified as :ig1-ig3
Incorrect Classification :(
Accuracy
Number of Images
Figure 4
Test Image from :dg1-dg5
Classified as :dg1-dg5
Correct Classification :)
Accuracy
Number of Images
Figure 5
Test Image from :gg1-gg3
Classified as :gg1-gg3
Correct Classification :)
Accuracy
Number of Images
Figure 6
Test Image from :hg1-hg3
Classified as :hg1-hg3
Correct Classification :)
Accuracy
Number of Images
Figure 7
Test Image from :ig1-ig3
Classified as :ig1-ig3
Correct Classification :)
Accuracy
Number of Images
Figure 8
Test Image from :jg1-jg3
Classified as :jg1-jg3
Correct Classification :)
Accuracy
Number of Images
Figure 9
Test Image from :kg1-kg3
Classified as :kg1-kg3
Correct Classification :)
Accuracy

Figure 10
File Edit View Insert Tools Desktop Window Help
Test Image from :lg1-lg10
Classified as :lg1-lg10
Correct Classification :)
Accuracy
Number of Images

Conclusion

The classifier classified the test leaves with 90% accuracy. More leaves are needed for each leaf type to get a better feature representation and thus increase accuracy.

Alternative Approach

One of the approach that we were planning to take was to get the SIFT [David Lowe et al] keypoint detectors for each class and then match the leaves using these descriptors. We have some preliminary results for those but due to the time constraints we have not implemented it completely. These features will be invariant to scale, rotation, translation and also to the affine transformation.